Hypopigmentation

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Introduction

- ☑ Hypopigmentation is one of the most common dermatological concerns in practice. It relates to decreased pigmentation. The major color determinant of our skin is melanin (eumelanin + pheomelanin) with a little contribution from ingested carotenoids and hemoglobin. MSH is the major hormone influencing melanin pigmentation.
- ☑ The cosmetic implication and social stigma attached to a hypopigmented patch in a dark-skinned person creates a lot of anxiety in the family.

Hypopigmentation is of two types:

1. *Melanopenic:*
   i. Anatomic defect (absent or damaged melanocytes), e.g., postinflammatory or post-chemical induced
   ii. Functional defect (defective tyrosine metabolism), e.g., pityriasis versicolor
2. Nonmelanopenic (aberrant neurovasculature response), e.g., nevus anemicus

Common Causes of Hypopigmentation

- ☑ *Congenital:*
  - Pigmentary mosaicism
- ☑ Post-traumatic
- ☑ Postinflammatory
- ☑ *Infective:*
  - Fungus (*Malassezia* species—pityriasis versicolor)
  - Bacteria (*Mycobacterium leprae*—Hansen's, *Cutibacterium acnes*—progressive macular hypomelanosis)
  - Viral (eruptive hypomelanosis—postviral exanthem)
  - Parasitic (post kala-azar)
- ☑ Neoplastic (T-cell—mycosis fungoides)
- ☑ Nutritional [kwashiorkor, copper, and selenium deficiency—malabsorption syndromes and total parenteral nutrition (TPN)]
- ☑ Miscellaneous (chemicals—skin-bleaching agents and lead-based cosmetics)
- ☑ Drugs [protein disulfide isomerase (PDI) inhibitors and steroids]
- ☑ Endocrine—hypopituitarism

Clinical Assist

- ☑ *Dermascope (an otoscope or a magnifying glass can substitute)—for better visualization of the lesion:*
  - Border (distinct in pityriasis versicolor) (indistinct in pityriasis alba)
  - Surface (scaling—pityriasis alba and pityriasis versicolor)
  - Atrophy (mycosis fungoides)
- ☑ *Wood's light:*
  - To differentiate hypopigmented from depigmented lesions
  - To appreciate hypopigmented lesions in light-colored individuals and infancy
  - Surface fluorescence—green-yellow fluorescence in pityriasis versicolor lesions
- ☑ Skin scraping (KOH preparation to appreciate fungal elements, e.g., spaghetti meat ball appearance in pityriasis versicolor) and in difficult cases
- ☑ Skin biopsy

They can be divided into:

☑ *With scaling:*

Diagnosis	*Features*	*Treatment*
Pityriasis alba	☑ May be associated with other atopic features ☑ Indistinct borders ☑ Asymptomatic	☑ Topical emollients ☑ 1% hydrocortisone cream (short duration) ☑ Reassurance about benign nature of lesion
Pityriasis versicolor	☑ Distinct borders ☑ Variable colors ☑ Asymptomatic ☑ Yellow-green fluorescence under Wood's light	☑ Any azole for local application twice a day for 4–6 weeks ☑ Oral fluconazole 8–12 mg/kg single dose repeated after a week in case of extensive lesions ☑ Ketoconazole lotion or selenium sulfide can be applied to lesions with short contact time of 10 minutes and then washed off
Seborrheic dermatitis	☑ Accompanied by dandruff	☑ 1% hydrocortisone cream twice a day for the lesion ☑ For the dandruff ☑ Ketoconazole shampoo or ☑ Zinc pyrethrum shampoo
Polymorphous light eruptions	☑ Light exposed areas ☑ Burning itching	☑ 1% hydrocortisone or desonide cream for the lesions twice a day ☑ Sun protection with sunscreens

☑ *Without scaling:*

Hansen's disease	☑ Sensory and sweating aberrations ☑ History of contact with an index case	Refer to a dermatologist
Early vitiligo	☑ The usefulness of Wood's light ☑ Sensations intact ☑ Hair color changes may be there ☑ Transition from hypopigmented to depigmented lesion	☑ Better to refer to a dermatologist ☑ *Treatments used:* – Steroid – Calcineurin inhibitors – Narrow-band ultraviolet B (UVB) – Psoralen and ultraviolet A (PUVA) therapy – Immunomodulators ☑ Skin grafting
Post-inflammatory	☑ Post-impetigo ☑ Post-procedure cryotherapy	☑ Reassurance ☑ Emollients
Nevus	☑ They grow with the body growth ☑ Achromicus with serrated border ☑ Anemicus with smooth border	☑ Reassurance ☑ Camouflaging agents
Contact with topical depigmenting agents		

Approach to a Hypopigmented Lesion on the Face

☑ *Without scaling and with organ involvement:*

Hypomelanosis of Ito	☑ Since birth ☑ Bizarre hypopigmented patches following Blaschko's lines ☑ May fade as they grow ☑ With ocular central nervous system (CNS) and skeletal involvement	Treat organ dysfunction
Tuberous sclerosis complex	☑ Ash-leaf macules ☑ Elliptical or lance-shaped patches ☑ Seen in infancy ☑ Presence of three or more lesions a major feature in diagnosis ☑ Seen in >90% of tuberous sclerosis complex (TSC) patients	Treat organ dysfunction

Recently Described New Entities of Hypopigmentation

☑ *Eruptive hypomelanosis:* Following a viral episode (post-viral exanthem—self-limiting)

☑ *Progressive macular hypomelanosis:* More on the trunk and upper extremities with a linkage to *Cutibacterium acnes* seen more in adolescents)

Condition	***Morphology***
Postinflammatory hypopigmentation	Depigmented patch/macule after resolution of primary insult (e.g., trauma, eczema, impetigo)
Tinea versicolor	Hypopigmented patches of scaly, coalescing macules, confirmed by potassium hydroxide of scrapings
Pityiasis alba	Appears on the face or upper arms as pale, slightly-scaly macules with indistinct borders
Vitiligo	Sharply-defined, milky-white acquired macules, typically adhering to a symmetrical and bilateral or segmental and unilateral distribution
Lichen striatus	Primary lesions linear, small, flat, skin-colored papules. Resolution can leave behind hypopigmented
Nevus anemicus	Congenital singular or clustered hypopigmented macules with well-defined, irregular borders typically found on the trunk. Macules will not redden when traumatized or exposed to heat
Nevus depigmentosus	Congenital singular hypopigmented macule patch with well-defined borders can present in a band-shaped pattern or extensive whorls. Will redden when stroked, unlike nevus anemicus
Tuberous sclerosis	Congenital ovoid (ash-leaf) dot-like (confetti) hypopigmented macules, associated with mental retardation and seizures
Mycosis fungoides	Pruritic ill-defined hypopigmented macules and patches on the trunk, buttocks, and extremities. Has a distinct fungal appearance that does not respond to antifungals
Piebaldism	Extensive, symmetrically distributed depigmentation on the forehead, front of thorax, and extremities, often by a midline lock of stark-white hair

- ☑ The diagnosis of hypopigmented disorders is in many cases made on clinical grounds, based on detailed medical history and physical examination.
- ☑ The lesion morphology, color (hypopigmented or depigmented), distribution, and pattern are important clues to diagnosis.
- ☑ However examination with a Wood's light, a skin biopsy and additional laboratory tests may be required if clinical diagnosis is uncertain.

Acquired hypopigmented disorders can be localized or diffuse.

After Wood's lamp examination they can be further subdivided as:

- ☑ Complete depigmentation
- ☑ Hypopigmentation
- ☑ Nonmelanotic leukoderma

Localized complete depig-mentation	***Localized hypopig-mentation***	***Localized nonmelanotic leukoderma***	***Diffuse complete depig-mentation***	***Diffuse hypopig-mentation***	***Diffuse non-melanotic leukoderma***
Vitiligo	☑ Postinflammatory ☑ Psoriasis ☑ Atopic dermatitis ☑ Contact dermatitis ☑ Pityriasis alba	Nevus anemicus	Generalized vitiligo	Severe malnutrition	Fair skin complexion
Halo nevus	☑ Infectious pityriasis versicolor ☑ Leprosy	Biar spots	Chemical leukoderma	Endocrine disorders	Anemia
Leukoderma punctata	☑ Drug-induced ☑ Topical/intralesional steroids ☑ Topical imiquimod		Melanoma-associated leukoderma	Idiopathic guttate hypo-melanosis	Skin edema
Melanoma-associated leukoderma	Nevus depigmentation			Leukoderma punctata	

Treatment/Management

The management of hypopigmented macules depends on the underlying pathophysiology. Successful repigmentation is often possible with timely diagnosis, removal of offending agents (chemicals and infectious agents), avoiding exposure to sunlight, using sunscreen, and appropriate therapeutic treatment. Repigmentation might not be possible in the congenital/inherited conditions (associated with chromosomal defects). Therapeutic treatment helps to fasten the process of repigmentation. Repigmentation is achieved by medications, phototherapy, and surgical procedures.

Medications

There are topical and systemic medications.

- ☑ *Topical corticosteroids:* Low-dose corticosteroids are used as first-line drugs in many hypopigmented conditions. They are known to accelerate the repigmentation process. Their use, along with phototherapy, also has great outcomes. Systemic corticosteroids are rarely used to treat hyperpigmentation, e.g., in vitiligo, to halt the rapid progression of skin lesions.
- ☑ *Topical calcineurin inhibitors:* These are topical immunomodulatory agents (tacrolimus and pimecrolimus), also used as first-line medication. Tacrolimus inhibits the synthesis and release of proinflammatory cytokines, thereby protecting melanocytes from the effects of T-cell and mast-cells. In contrast to topical steroids, tacrolimus does not cause skin atrophy and striae, preferred for treating facial hypomelanosis conditions.
- ☑ *Vitamin D:* The current literature reviews that vitamin D has an association with skin pigmentation. Vitamin D increases melanogenesis by increasing the tyrosinase content in melanocytes by the antiapoptotic effect.
- ☑ *Antifungals:* These are used in treating tinea versicolor infection, available in both topical and systemic forms. Topical agents are often used for weeks to months; include selenium sulfide shampoo, 1% or 2% ketoconazole ointment, and zinc pyrithione. Oral medications such as fluconazole and itraconazole are used for a short duration of therapy.
- ☑ *Oral isotretinoin:* This is used to treat conditions like progressive macular hypomelanosis (PMH) in which *Propionibacterium acnes* is the underlying cause.
- ☑ *Emollients:* Used to provide a soothing effect in pruritic and scaly skin lesions.

Treatment/Management

Phototherapy

- ☑ This includes narrow-band ultraviolet B (NBUVB) and psoralen and ultraviolet A (PUVA).
- ☑ NBUVB is superior and preferred over PUVA in treating vitiligo.
- ☑ PUVA has comparatively more adverse effects because of the systemic use of psoralen. PUVA is contraindicated in children and pregnant women.

Surgical Procedures

Skin grafting and split skin grafting are practiced in treating some localized depigmented and inherited hypopigmented conditions.

Vitiligo Special Considerations for Treatment

- ☑ Localized areas of vitiligo may respond to potent topical steroid, topical tacrolimus, or topical pimecrolimus.
- ☑ In patients with more extensive involvement, NBUVB (UVB 311) is the treatment of choice.
- ☑ Systemic therapy and whole body depigmentation is rarely used in children.
- ☑ In all forms of vitiligo, response to therapy is slow taking many months to years.
- ☑ For those not interested in treatment cover up cosmetics may be used.
- ☑ All areas of vitiligo are susceptible to sun damage and care should be taken to minimize sun exposure of affected areas.
- ☑ Spontaneous remission may be seen in small percentage of cases.

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Dengue in Children

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